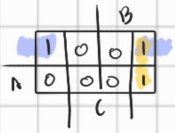
$Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C} =$

$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}B\bar{C} + AB\bar{C} =$

$= \bar{A}\bar{C}(\overset{1}{\bar{B}+B}) + B\bar{C}(\overset{1}{\bar{A}+A}) =$

$= \underbrace{\bar{A}\bar{C} + B\bar{C}}_{si} = \cancel{\bar{C}(\underset{NO}{\bar{A}+\bar{B}})}$



$\rightarrow \bar{A}\bar{C} + B\bar{C}$

$Y = \bar{A}\bar{B}\bar{C} + \overset{or}{\bar{A}B\bar{C}} + \overline{AB\bar{C}}_{and}$

Leggi di Morgan

$\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A} + \bar{B} + C$

$\bar{A}\bar{C}(\bar{B}+B) + \bar{A} + \bar{B} + C$

$\bar{A}\bar{C} + \bar{A} + \bar{B} + C$

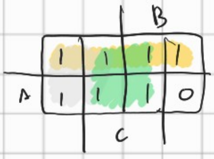
$\bar{A}(\underline{\bar{C}+1}) + \bar{B} + C$

$\downarrow 1$

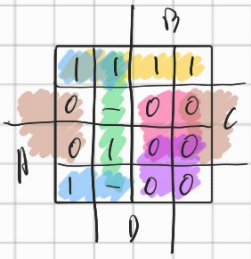
$\bar{A} + \bar{B} + C \leftarrow$ questa è la sintesi minima

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

$\Rightarrow$



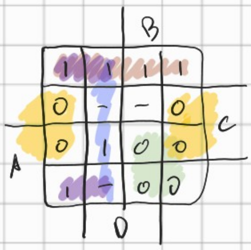
$\underline{\bar{A} + \bar{B} + C}$



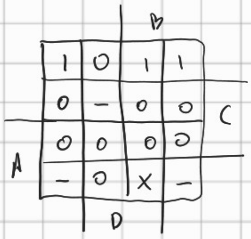
$\bar{A}\bar{C} + \bar{B}\bar{C} + \bar{B}D$

$(\bar{A}+\bar{B}) \cdot (D+\bar{C}) \cdot (\bar{C}+\bar{B})$

$\neq$

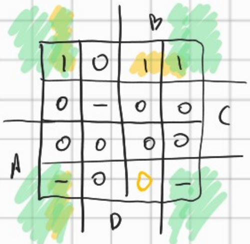


KMap Variabili riportate



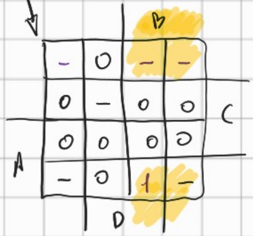
V.R. è X

X=0



$\bar{C}\bar{D} \quad + \bar{A}B\bar{C}$

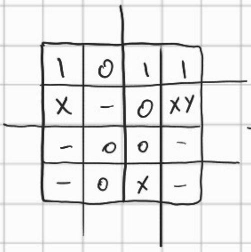
X=1



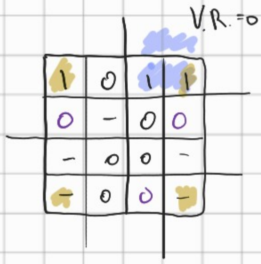
$B\bar{C}$

$\Longrightarrow T_{OT} = \bar{C}\bar{D} + \bar{A}B\bar{C} + X(B\bar{C})$

V.R. ~~sono~~ X , XY

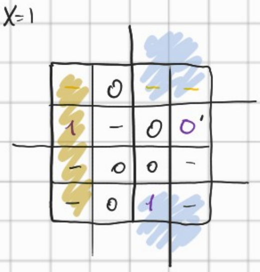


V.R. = 0



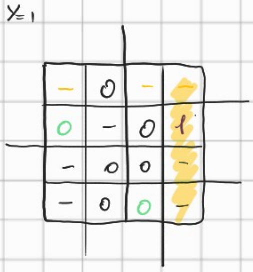
$\underline{AB\bar{C} + \bar{C}\bar{D}}$

X=1



$\bar{B}\bar{D} + B\bar{C}$

XY=1



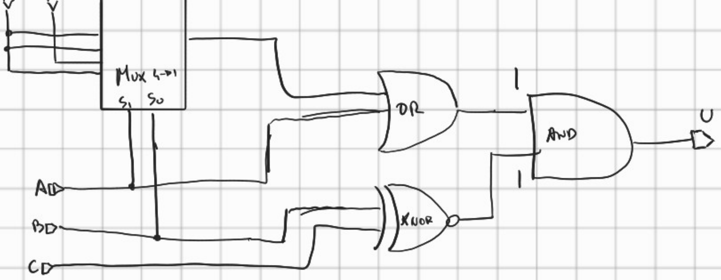
$B\bar{D}$

$T_{OT} = \bar{A}B\bar{C} + \bar{C}\bar{D} + X(\bar{B}\bar{D} + B\bar{C}) + XY(B\bar{D})$

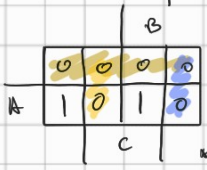
1 Uscita

3 ingressi

A	B	C	U
0	0	0	0
0	0	1	0
0			0



	1	0	
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



0	0	0	0
1	0	1	0

AND-OR → $A\bar{B}\bar{C} + ABC$

OR-AND → $A \cdot (B+\bar{C}) \cdot (\bar{B}+C)$

Basi e operazioni binarie
(da fare con procedimento)

$11_{10} \to$ BIN

$11 \,|\, 2 \to 1$
$5 \,|\, 2 \to 1$
$2 \,|\, 2 \to 0$
$1 \,|\, 2 \to 1$
0

$\overset{2^3\,2^2\,2^1\,2^0}{1011_2} = 11_{10}$

$10_{10} \to$ OTTALE

$10 \,|\, 8 \to 2$
$1 \,|\, 8 \to 1$
0

$\to 12_8 = 10_{10}$

$30_{10} \to$ HEX

$30 \,|\, 16 \to 14 \to E$
$1 \,|\, 16 \to 1$
0

↑ $\underline{1E_{16}}$

Con la conversione negativa non devo fare l'es. inverso

$-11_{10} \to$ complemento a 2 a 8 bit

$\underline{1011}$

→ estendo a 8 bit

00001011
↓ NEGO
11110100
→ sommo 1 → 11110101_2

$$\begin{array}{ccccccccc}
 & +1 & +1 & & & +1 & & & \\
 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & + \\
+1 & & 1 & 1 & 0 & 0 & 1 & 0 & = \\
\hline
1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 &
\end{array}$$

$$\begin{array}{cccccccc}
\overset{0}{\cancel{1}} & 0 & 1 & 0 & \overset{0}{\cancel{1}} & \overset{1}{0} & 0 & - \\
 & 1 & 1 & 0 & 0 & 0 & 1 & = \\
\hline
0 & 1 & 0 & 0 & 0 & 1 & 1 &
\end{array}$$

$$\begin{array}{cccccccc}
\overset{0}{\cancel{1}} & \overset{1}{0} & \overset{0}{\cancel{1}} & \overset{1}{0} & \overset{0}{\cancel{1}} & \overset{1}{0} & 0 & - \\
 & 1 & 1 & 1 & 1 & 1 & 1 & = \\
\hline
0 & 0 & 1 & 0 & 1 & 0 & 1 &
\end{array}$$

$$\begin{array}{ccccccc}
 & & 1 & 0 & 1 & 0 & \cdot \\
 & & & 1 & 1 & 0 & = \\
\hline\hline
 & & & & & - & + \\
 & 1 & 0 & 1 & 0 & - & + \\
1 & 0 & 1 & 0 & - & - & = \\
\hline
1 & 1 & 1 & 1 & 0 & 0 &
\end{array}$$

$$\begin{array}{cccccccccc}
 & & & & & 1 & 0 & 1 & 0 & \cdot \\
 & & & & 1 & 0 & 1 & 1 & 0 & = \\
\hline
 & & & & & & & & X & + \\
 & & +1 & \overset{+1}{1} & \overset{+1}{0} & 1 & 1 & 0 & - & + \\
 & & 1 & 0 & 1 & 1 & 0 & - & - & + \\
 & +1 & & & & & & & X & + \\
1 & 0 & 1 & 1 & 0 & - & - & - & - & + \\
\hline
 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 &
\end{array}$$

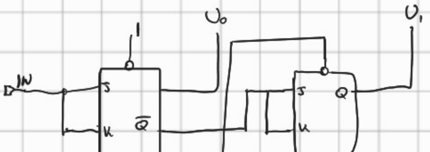
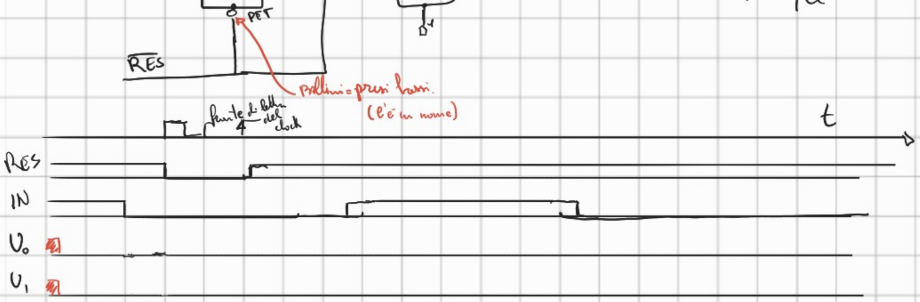


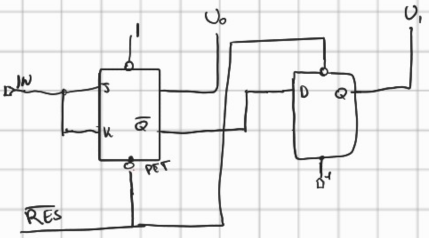
Tavola di Eccitazione

S K	Q
0 0	Q
0 1	0
1 0	1
1 1	Q



Pallini = presi bassi. (l'è un nome)

fronte di salita del clock



D	Q
0	0
1	1